

Marion Fumani Maphali

Network Engineer · Backend Developer · IT Professional

marionmaphalif@gmail.com

+27 61 124 2770

[View LinkedIn Profile](#)

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4638 Golf Course Flats, Pimville
Zone 4, Soweto, Johannesburg,
SA

PROFESSIONAL SUMMARY

IT Communication Networks graduate from Nelson Mandela University with hands-on experience in enterprise network design, Cisco routing and switching, and Python backend development. Completed a 12-month backend developer internship at DataTank BI Solutions, building data-driven systems within a professional BI environment. Holder of multiple Cisco Networking Academy certifications covering enterprise networking, cybersecurity, IoT, and enterprise automation and Security. Driven by a passion for continuous learning, technical problem-solving, and building reliable infrastructure.

PROFESSIONAL EXPERIENCE

Python Backend Developer Intern

2024 – 2025

DataTank BI Solutions · North Riding, Randburg

Hosted by MICTSETA with Luphindi Business Solutions | 12-Month Internship

- Developed backend software systems using Python for business intelligence applications in a live enterprise environment.
- Built and maintained database interactions and backend business logic for BI data processing pipelines.
- Provided system support and implemented technical solutions across multiple BI projects.
- Collaborated with a cross-functional technical team on software development tasks, debugging, and optimization.
- Applied analytical thinking to design reliable, efficient backend systems aligned with client requirements.

Network Design Engineer (Academic Project)

2024 – 2024

Nelson Mandela University · Port Elizabeth, South Africa

IT Communication Networks Capstone Project | 12-Month Academic Project

- Designed and implemented a full-scale enterprise airport network infrastructure as part of the final-year capstone project.
- Applied Cisco routing and switching technologies (OSPF, VLANs, BGP, ACLs) across a complex multi-zone topology.
- Implemented VPN, IPSec, NAT/PAT, firewall rules, and network segmentation for security and traffic optimization.
- Produced comprehensive technical documentation, network diagrams, and architecture specifications.
- Simulated and validated the network design using Cisco Packet Tracer and virtual machine environments.

EDUCATION

Diploma in Information Technology: Communication Networks

2021 – 2025

Nelson Mandela University (NMU) · Port Elizabeth

Graduated: April 2025 · Specialization: IT Communication Networks

National Senior Certificate (NSC / Matric)

2020

E.P.P. Mhinga High School · Limpopo

Key subjects: Mathematics, Life Sciences, Physical Sciences, Accounting, Xitsonga, English

CERTIFICATIONS

CCNA: Enterprise Networking, Security & Automation	Cisco NetAcad / NMU	<i>Jan 2025</i>
CCNAv7: Introduction to Networks	Cisco Networking Academy	<i>Dec 2021</i>
Introduction to Cybersecurity	Cisco Networking Academy	<i>Aug 2021</i>
Introduction to IoT	Cisco Networking Academy	<i>Jul 2022</i>
Diploma in IT: Communication Networks	Nelson Mandela University	<i>Apr 2025</i>
National Senior Certificate	Dept. of Basic Education, SA	<i>Nov 2020</i>

TECHNICAL SKILLS

Networking Protocols: TCP/IP, OSI model, IPv4/IPv6, Subnetting, LAN, WAN, VLAN, WLAN

Routing & Switching: Static Routing, OSPF, BGP, EIGRP, VLANs, STP, Cisco IOS

Network Security: Cisco ASA Firewall, VPN (IPSec, Site-to-Site), ACLs, IDS/IPS, NAT/PAT

Tools & Platforms: Cisco Packet Tracer, PuTTY, VirtualBox, Azure (Networking basics)

Programming: Python (backend scripting & automation), C#, SQL

Troubleshooting: SSH/Telnet, Ping, Traceroute, Wireshark

Operating Systems: Windows (advanced), Linux (CLI basics)

KEY PROJECTS

Airport Network Infrastructure Design & Implementation

Designed a complete enterprise-grade airport network as a capstone project, covering full topology planning, VLAN segmentation, routing protocol configuration (OSPF, BGP), VPN setup, and security hardening across a multi-zone environment using Cisco IOS and Packet Tracer.

Python BI Backend Systems · DataTank BI Solutions

Built and maintained Python backend systems for business intelligence reporting, including database query logic, data processing pipelines, and backend APIs during a 12-month professional internship.

Personal Portfolio Website

Designed and built a fully responsive personal portfolio website using HTML5, CSS3, and vanilla JavaScript, featuring animated network node canvas, scroll-triggered transitions, and a contact form. Deployed on GitHub Pages.

ADDITIONAL INFORMATION

Languages: Xitsonga (Native), English (Fluent)

Driver's Licence: Code 10 / Class C1 (Valid until 2030)

Portfolio: marion-mars.github.io/marion-maphali-portfolio

Availability: Open to opportunities — immediate start